



Introduction

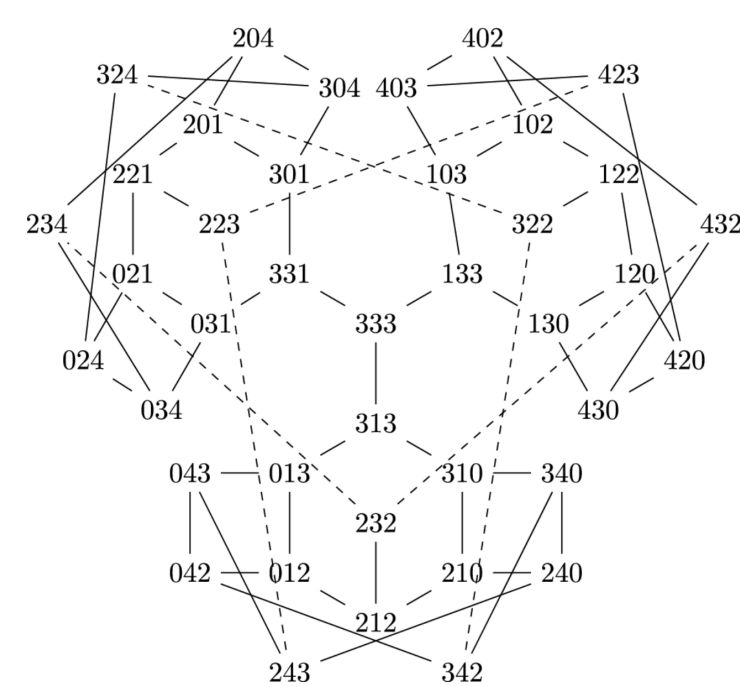
Goal: Our group's goal is to show that the solutions to the Markoff equation $x^2 + y^2 + z^2 = xyz \pmod p$ are connected for all primes p . We have been working through Daniel Martin's paper on this topic where he turns the problem into a linear algebra span problem. His solution is a computational algorithm which can tell us for a given prime if the graph is connected $\pmod p$. Our research worked to make minor optimizations to Martin's algorithm in order to increase the lower bound of connectivity that he provided.

Background

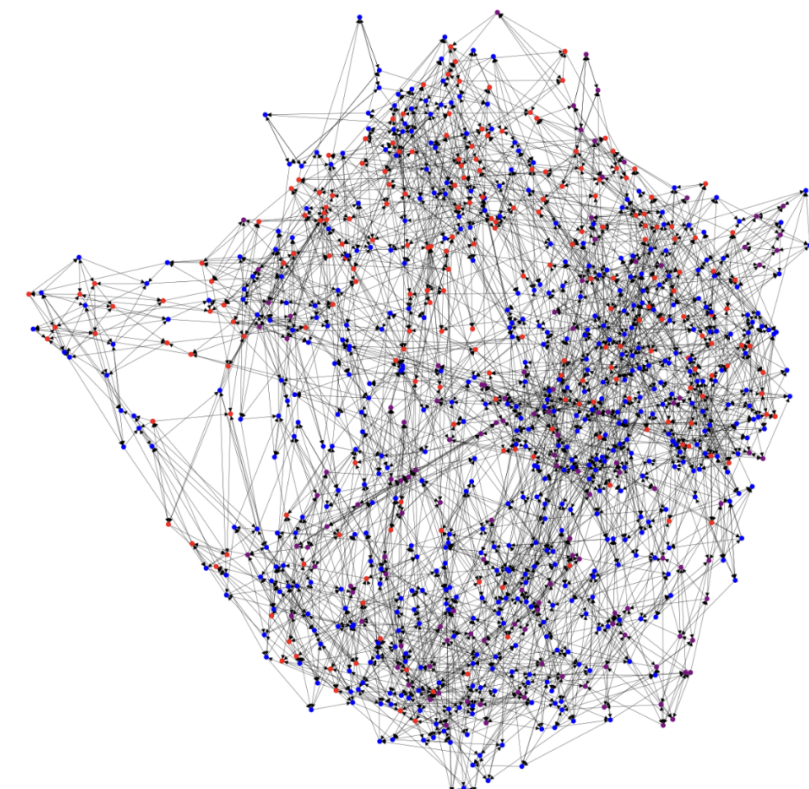
The foundation of our research is the Markoff Tree; the set of all integer solutions to the Markoff equation. The solutions form nodes of an infinite tree $\mathcal{G}$ with edges that are transformations of the group Γ . This group is generated by Vieta involutions (fixing two coordinates of a known solution and solving for the third), coordinate permutations, and sign swaps. If we reduce this graph $\pmod p$ we then get the Markoff graph $\pmod p \mathcal{G}_p$. We want to show that this graph is connected, ie given any $v, w \in \mathcal{G}_p$ there exists a $\gamma \in \Gamma$ such that $\gamma(v) = w$. Martin's approach to showing this transitivity is framed through the Q classification conjecture.

Conjecture. (*Q Classification Conjecture*) *The markoff class of essential triples is uniquely determined by $\kappa := x^2 + y^2 + z^2 - xyz$.*

This means that all essential triples are located in the same orbit under the action of Γ .



Markoff Mod 5 Graph



Markoff Mod 31 Graph

In order to show that the Q Classification Conjecture holds we must make a computational concession.

Let $\mathcal{O} \subseteq \mathcal{G}_p$ be any connected component of $\mathcal{G}_p$, $C_{\mathcal{O}}(\alpha) = |\{(\alpha, \beta, \gamma) \in \mathcal{O}\}|$ and

$$V_{\mathcal{O}} = \begin{pmatrix} C_{\mathcal{O}}(1) \\ C_{\mathcal{O}}(2) \\ \vdots \\ C_{\mathcal{O}}(p) \end{pmatrix} \in \mathbb{F}_p^p. \text{ The general idea of the construction is that the number of}$$

orbits corresponds to the number of $V_{\mathcal{O}}$ by construction. So we need to prove that the span of these vectors $V_{\mathcal{O}}$ is sufficiently small to prove that there are a sufficiently small number of orbits, ie connectivity. Martin does this cleverly by defining a set of polynomials which vanish when summed over each orbit.

Let R be an integral domain and $\bar{R}$ be the integral closure of R in its fraction field, now $\mathcal{P}(\mathbb{F}_p) := \{f \in \bar{R}[x] : \sum_{t \in \mathcal{O}} f(t) = 0\}$. Now by a simple reordering argument we have that for a given $f \in \mathcal{P}(\mathbb{F}_p)$. (*Next column*)

Background

$$0 = \sum_{t \in \mathcal{O}} f(t) = \sum_{\alpha \in \mathbb{F}_p} f(\alpha) C_{\mathcal{O}}(\alpha) = \begin{pmatrix} f(1) \\ \vdots \\ f(\alpha) \end{pmatrix} \cdot \begin{pmatrix} C_{\mathcal{O}}(1) \\ \vdots \\ C_{\mathcal{O}}(p) \end{pmatrix}, \text{ so the vectors are orthogonal}$$

and $\mathcal{P}(\mathbb{F}_p)$ is a subspace, this means that $\dim(\mathcal{P}(\mathbb{F}_p)) + \dim(\mathcal{P}^{\perp}(\mathbb{F}_p)) = \dim(\mathbb{F}_p^p) = p$. Now by construction, $f \in \mathcal{P}(\mathbb{F}_p)$ if and only if the dot product is equal to zero, so the goal of the algorithm is to produce a sufficient amount of polynomials in $\mathcal{P}(\mathbb{F}_p)$, in order to show the dimension of $\mathcal{P}^{\perp}(\mathbb{F}_p)$ is small and the less connected components we have in our graph.

Main Results

Theorem (Main Theorem). *For every prime $p < 2.43 \times 10^{13}$, the Markoff mod p graph G_p is connected. Here the vertices of G_p are the nonzero solutions of $x^2 + y^2 + z^2 \equiv xyz \pmod p$, and the edges are the Vieta involutions R_1, R_2, R_3 .*

Connectivity of G_p known for	Source
all but finitely many p	Bourgain–Gamburd–Sarnak, Chen
$p > 3.448 \cdot 10^{392}$	Eddy–Fuchs–Litman–Martin–Tripeny
$p < 10^6$	Brown
$2 \operatorname{lcm}(1, \dots, 17) \nmid p^2 - 1$ ($p < 388,961$)	Martin
$p < 2.43 \cdot 10^{13}$	Our work

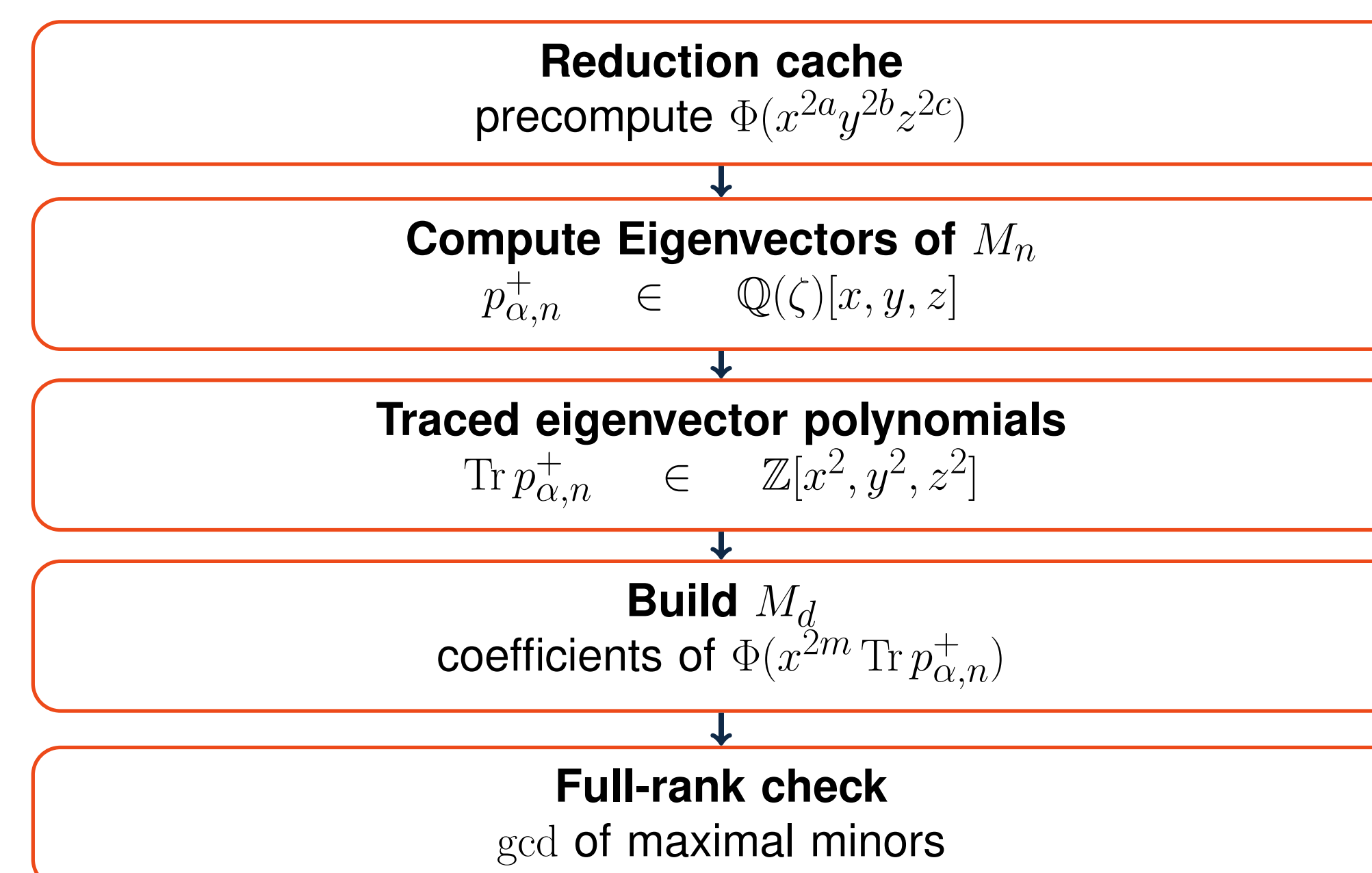
Martin's Criterion

For a prime power d , Martin builds an integer matrix M_d whose columns come from polynomials that sum to zero over Markoff orbits. Those polynomials are the eigenvectors of a matrix M_n , given explicitly in Martin's paper: If

$\operatorname{reduce_int}(\gcd(\det M_1, \dots, \det M_k)) = 1$, where $M_1, \dots, M_k$ are the maximal minors of M_d , then M_d has full rank $\pmod p$ every $p \geq 2n_d$. This gives connectivity for all $p \not\equiv \pm 1 \pmod{2d}$. Verifying every prime power $d \leq D$ then proves G_p is connected whenever

$$2 \operatorname{lcm}(1, 2, \dots, D) \nmid p^2 - 1.$$

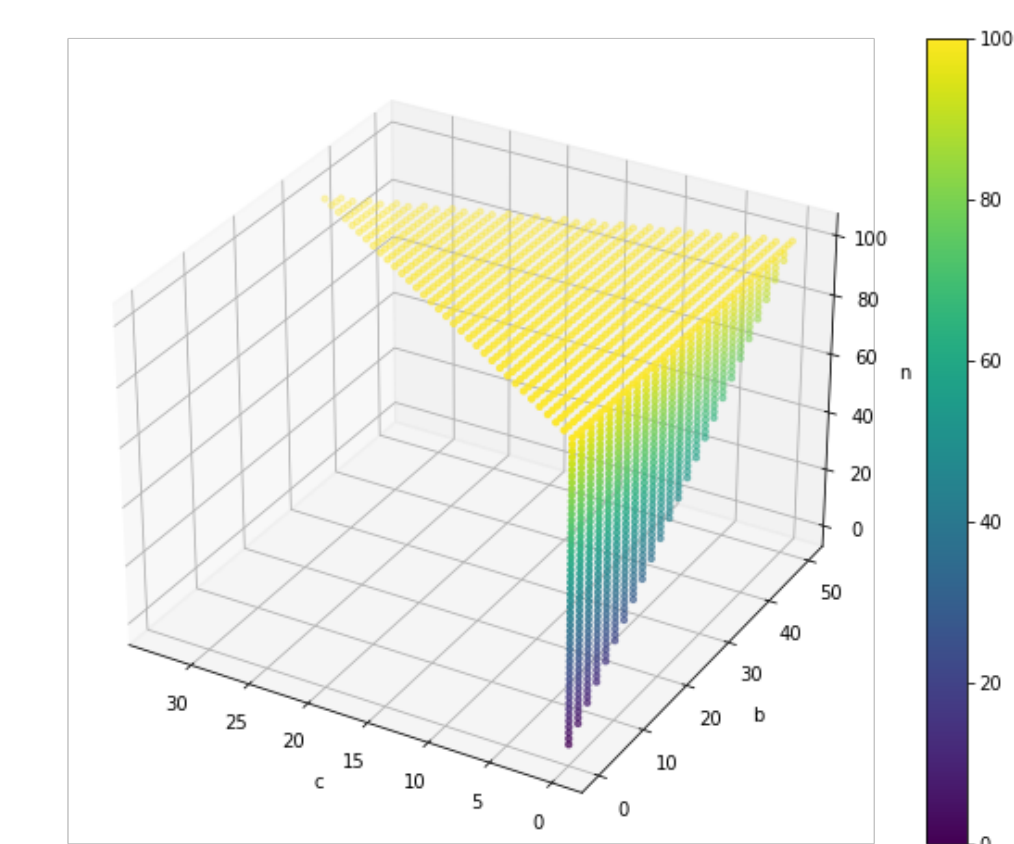
The Pipeline



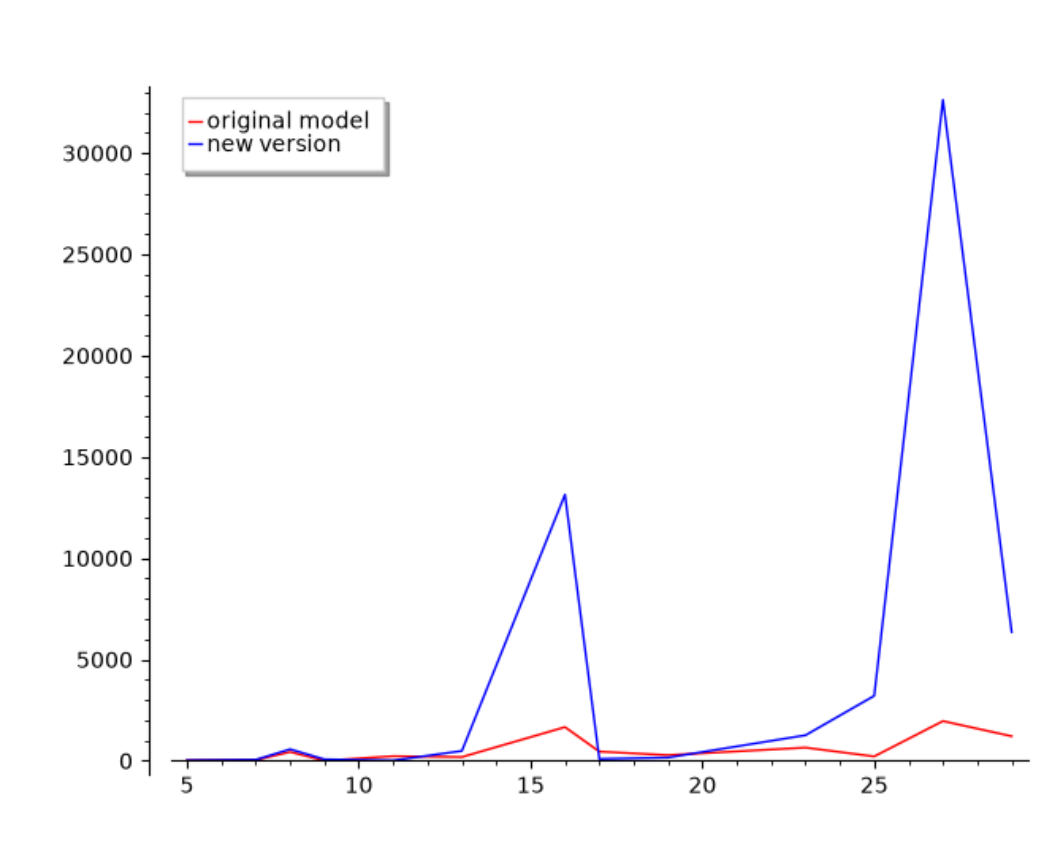
Our Changes

Main Change: Improvement of Reduction Cache

Initially we needed to store most of the space, but we later found that many coordinates were actually not needed, so we deleted these redundant parts. The final storage method is shown in the figure, we only stored information of two 2-dimensional planes (Graph 1).



Graph 1



Graph 2

Additional Failed Attempts

1. We tried to avoid computing all the eigenvectors directly and instead using old eigenvectors to generate new ones via symmetry. This would've allowed us to store lower degree polynomials in the reduction cache. However, it ended up being much slower than the original version (Graph 2), due to us needing to perform Gaussian elimination over a field extension.
2. We tried to use SNF form to check whether the matrix is full rank. Unfortunately, the degrees of polynomials grow exponentially when performing SNF operations, so for large matrices this would cause an overflow.

Future Directions

An important corollary:

Notation 6.18. For $n \geq 0$, let

$$f_n(x^2, y^2) = \sum_{j=0}^n \frac{2n}{n+j} \binom{n+j}{n-j} (x^2 - 4)^j (\kappa - x^2)^{n-j} y^{2j}.$$

Corollary 6.19. For any integers $d, n \geq 2$ and any $g(x^2) \in \bar{R}[x], g f_n \in \mathcal{P}_x(R, d)$. Furthermore,

$$\Phi(g f_n) = \frac{1}{n} \sum_{j=0}^n \sum_{\alpha \in \hat{\Lambda}_n} \binom{2n-j-1}{j} g(\alpha^2) (\alpha^2 - 4)^{n-j} (\kappa - \alpha^2)^j b_{n-j}(x^2) + r(x^2)$$

for some $r(x^2) \in \bar{R}[x]$ of degree at most $2 \lfloor \frac{3}{4}n \rfloor$.

This Corollary implies that if we can find a specific equation for $r(x^2)$, then we can repeatedly apply it to the remainder term. This would allow us to eliminate the reduction cache altogether, removing the primary space bottleneck.

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